

Final Exam Classification

Methodology & Experimental Design

Preprocessing & Leakage

Data Imputation: Dealt with the missing values in agent, company, and country using median/constant strategies.

Leakage Prevention: The columns of 'reservation status' and the dates have been removed; the columns are only filled out after the occurrence.

Standardization: Constructed a pipeline to do One-Hot Encoding and Numeric Scaling to achieve feature parity.

Model Framework

70/15/15 Split: Split by partitioning 119k observations into Training, Validation and objective Test sets.

Algorithms: Benchmarked Logistic Regression, DT, RF, GBM, KNN, and Polynomial SVC.

Assessment: Prefer Recall and F1-Score over simple Accuracy to assess business risk.

Speaker Script

In this project, we were able to analyze more than 119,000 hotel bookings to predict cancellations. A critical discovery was that we could hit fake 100 percent accuracy when predicting cancellations using just one column of our models, namely the reservation status column. If we had kept all of the data in every column of our models, then we would have used the critical path and identified the variables we cared about the most. A critical discovery was that we could hit fake 100 percent accuracy when predicting cancellations using just one column of our models, namely the reservation status column.

Model Performance Benchmarking

Model Algorithm	Val Acc	Val Prec	Val Recall	Val F1	Test AUC
Logistic Regression	0.804	0.812	0.591	0.684	0.865
Decision Tree	0.838	0.845	0.678	0.752	0.881
Random Forest	0.845	0.901	0.645	0.752	0.909
Gradient Boosting	0.858	0.865	0.724	0.788	0.919
K-Nearest Neighbors	0.818	0.786	0.692	0.736	0.868
Bayesian Stacking (Winner)	0.865	0.872	0.738	0.800	0.925

Speaker Script

To the results, which model performed best? Gradient Boosting, our best single model, but our custom Bayesian Stacking Ensemble was our best overall model. It actually learned which model to trust given specific guest profiles, pushing its F1-score to 0.80 and its AUC to 0.925. Notice that tree-based models outperformed distance-based models (such as KNN and SVC) significantly on this tabular data.

Business Impact & Recommendation

73.8% Recall

Accurately recognizes almost 3 out of 4 cancellations with aggressive reselling allowed.

Revenue Recovery

Minimizes the rate of the False Negative- the rooms will not have to sit in vain due to unexpected cancellations.

Actionable Lead Time

Forecasts the results at the beginning of booking cycle depending on the types of lead-time and deposit.

Conclusion: The best production model is the Bayesian Stacking Ensemble. It offers the best trade-off between accuracy and high-value recall that would be required in managing revenues.

Speaker Script

To sum up, we would like to speak about the so-called False Negative--that is, predicting that a guest will show up when he or she actually cancels the hotel. This will result in the empty rooms and loss of revenue. Our winner, the Bayesian Stacker, achieved a 73.8% Recall rate. This means that the hotel will be able to identify 73.8% of all cancellations in advance.